Project Documentation Phase

Project Report

Date	28-10-2026
Team ID	SWTID-2026-7875
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

Project Phases

1. Brainstorming & Ideation Phase

- **Objective:** Secure record access across multiple departments using the same table.
- **Core Problem:** Standard security allows cross-department users to view sensitive data outside their scope.
- **Solution:** Create script-controlled Access Control Lists (ACLs) using custom roles (bb1–bb4) to filter records dynamically based on field values (e.g., Branch = EEE).

2. Requirement Analysis Phase

- **Functional Requirements:**
  - Create test users (EEE User) and custom roles (bb1, bb2, bb3, bb4).
  - Create a custom table Institution Details (u_institution_details) with fields: Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, and Description.
  - Implement ACLs for Read (bb1), Create (bb2), Write (bb3), and Delete (bb4) operations.
- **Non-Functional Requirements:**
  - Follow least privilege access ("Deny Unless explicitly allowed").
  - Execute real-time security checks without UI performance lag.

3. Project Design Phase

Access Control Matrix

Operation	Required Role	Execution Condition / Script	Behavior
Read	bb1	Branch == 'EEE' + gs.hasRole('bb1')	Displays EEE branch records only.
Create	bb2	Role-based (bb2)	Enables the **New** button.
Write	bb3	Role-based (bb3)	Permits editing existing records.
Delete	bb4	Role-based (bb4)	Enables the **Delete** action.
All Ops	admin	System Admin Role Override	Full view and edit access across all records.

4. Project Planning Phase

- **Target Execution Time:** 3 Hours.
- **Implementation Steps:**
    1. User & Role Setup (30 mins)

2. Custom Table & Sample Data Setup (45 mins)
3. Read, Create, Write, and Delete ACL Configuration (60 mins)
4. Impersonation Testing & Verification (45 mins)

5. Project Development Phase

1. **Users & Roles:** Created EEE User and assigned custom roles bb1, bb2, bb3, and bb4.
2. **Table Creation:** Created u_institution_details and populated sample data for ECE, EEE, and CSE branches.
3. **Read ACL:** Added Branch is EEE data condition, assigned bb1 role, and configured script logic:JavaScript
4. **Operations ACLs:** Created Create (bb2), Write (bb3), and Delete (bb4) ACLs.

6. Project Testing Phase

Test Scenario	**Active User Roles**	**Target Data**	**Result**
Read Filtering	bb1	u_institution_details	**PASS:** Only EEE records visible.
Unauthorized Read	No bb1	u_institution_details	**PASS:** "No records to display".
Admin Access	admin	u_institution_details	**PASS:** All records (ECE, EEE, CSE) visible.

Record Operations	bb2, bb3, bb4	Record List & Forms	**PASS:** Create, Write, and Delete buttons enabled.

7. Project Documentation Phase

- **Application Scope:** Global
- **Update Set:** ACL Project (Global)
- **Security Rules:** 4 ACLs created combining user role verification and field-level condition evaluation.

8. Project Demonstration Phase

- **Read Enforcement:** EEE User with bb1 sees only INS0001007 and INS0001004 (EEE records).
- **Admin Verification:** Administrator bypasses branch filters to display all records.
- **Delete Execution:** EEE User with bb4 successfully selects and deletes records via list actions.